

3 Yr. Degree/4 Yr. Honours 6th Semester Examination, 2026 (CCFUP)

Subject : Chemistry

Course : CHEM 6013 (MAJOR)

(Physical Chemistry)

Time: 3 Hours

Full Marks: 60

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*

1. Answer any ten questions.

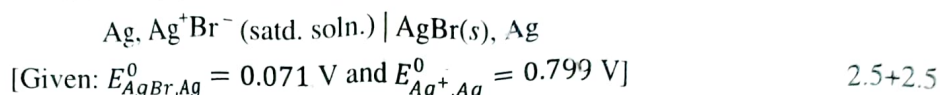
2×10=20

- (a) Prove that the elevation of boiling point (ΔT_b) of a dilute solution is a colligative property.
- (b) Draw the graphical plots of the chemical potential (μ) versus temperature (T) of solid, liquid and gaseous phases of a pure substance, and also comment on the slope of the plots. 1+1
- (c) $\text{NH}_4\text{Cl}(s)$ decomposes to $\text{NH}_3(g)$ and $\text{HCl}(g)$ on heating in an evacuated container. Find the number of components (C) and phases (P) in this system. 1+1
- (d) State the Nernst distribution law, and write down the conditions for the validity of the law.
- (e) Calculate the mean activity coefficient ($\gamma_{\pm}$) of 0.01 molal NaCl solution at 25°C.
[Given: Debye-Huckel constant (A) = 0.509]
- (f) From Clapeyron equation, explain how does boiling point vary with the variation of pressure.
- (g) What do you mean by reversible cell? Give an example of a reversible cell with cell reaction. 1+1
- (h) Draw the potentiometric titration curve obtained in case of an acid-base titration, and also comment on the utility of the curve. 1+1
- (i) Compare the values of the rotational constants (B) of hydrogen (H_2) and hydrogen deuteride (HD) and give reason in support of your answer.
- (j) How many vibrational degrees of freedom are there for CO_2 molecule? Name them and also state which are IR active and IR inactive.
- (k) Stokes' lines are more intense than anti-Stokes' lines.— Explain.
- (l) Why tetramethyl silane (TMS) is used as an universal internal reference standard in measuring the NMR spectrum?
- (m) Define the term 'molar extinction coefficient or molar absorptivity (ϵ)'. How can you find the value of ' ϵ ' for a coloured species experimentally? 1+1
- (n) Absorption and fluorescence hold a mirror image relationship — comment.
- (o) How azeotropic mixtures can be separated?

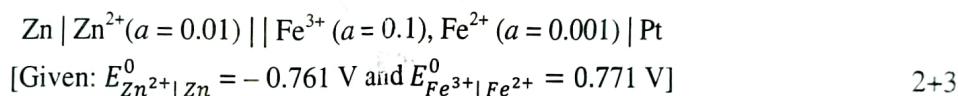
2. Answer any four questions.

5×4=20

- (a) (i) Explain the cause of lowering of vapour pressure of a solvent due to the addition of small amount of solute to have an ideal solution.
- (ii) Establish the mathematical relation between the osmotic pressure (π) and the depression of freezing point (ΔT_f) of a dilute solution. 2+3
- (b) (i) Derive the relation between the activity (a) and the mean activity coefficient ($\gamma_{\pm}$) of an electrolyte of the type M_xA_y if the molality = m .
- (ii) Calculate the solubility product of AgBr in water at 25°C from the cell:



- (c) (i) A quinhydrone electrode cannot be used at above the pH = 8. — Explain.
- (ii) Calculate the emf (E), the change in free energy (ΔG) and the value of equilibrium constant (K) for the cell reaction occurring in the cell:



- (d) (i) The force constant of a harmonic oscillator having reduced mass of $1.5 \times 10^{-27} \text{ kg}$ is given as 10 Nm^{-1} . Calculate the zero-point energy.
- (ii) What is Larmor frequency? Comment on its importance in NMR spectroscopy. 2+3
- (e) (i) What is the value of ' J ' in the most highly populated rotational level of ICl molecule [rotational constant (B) = 0.114 cm^{-1}] at room temperature?
- (ii) State the Einstein's law of photochemical equivalence. Also comment on its validity. 2+(1+2)

- (f) (i) How can you study the photochemical decomposition of oxalic acid using a chemical actinometer? [$(\text{COOH})_2 \rightarrow \text{CO}_2 + \text{CO} + \text{H}_2\text{O}$]

- (ii) Consider the photochemical reaction: $A \xrightarrow{h\nu} B + C$

When irradiated with light of wavelength $2537\text{\AA}$ for a certain period of time, the energy absorbed = $3.436 \times 10^8 \text{ erg}$, and the number of moles of B formed as one of the products = $3.64 \times 10^{-6} \text{ moles}$. Calculate the quantum yield of the given photochemical reaction. 2+3

3. Answer any two questions.

10×2=20

- (a) (i) An aqueous solution (5% by mass) of a non-volatile solute boils at 100.15°C . Calculate the molar mass of the solute. [Given: K_b for water = 0.52 kg mol^{-1}]
- (ii) Write down the Duhem-Margules equation. Hence prove that if one component shows positive deviation of vapour pressure the other component will also show positive deviation.
- (iii) Draw the phase diagram of water (H_2O) with proper labelling. Also calculate the number of degrees of freedom along any one curve drawn and at the triple point in the phase diagram of water (H_2O). 3+(1+2)+(2+2)

- (b) (i) What do you mean by concentration cell without transference? Give an example of such a cell with the cell diagram and also write down the expression for its emf.
- (ii) Starting from $\log \gamma_i = -BZ_i^2 \sqrt{\mu}$, arrive at the expression of $\log \gamma_{\pm}$ for the electrolyte $M_x N_y$. (Terms have their usual meaning.)
- (iii) Calculate the emf of the electrode-concentration cell:
 $\text{Pt}; \text{H}_2 (p_1), \text{HCl} (aq), \text{H}_2 (p_2); \text{Pt}$
 at 25°C if the values of p_1 and p_2 are 600 torr and 400 torr, respectively. (2+2)+3+3
- (c) (i) The nature of Raman spectrum of a substance depends on both the nature of molecules and the wavelength of incident radiation.— Justify or criticise.
- (ii) Using a 60 MHz instrument, the difference in frequency between the absorption by a proton and by TMS (reference standard) was found to be 430Hz. Calculate the chemical shift in ppm unit.
- (iii) The spacing between the rotational lines in the IR spectrum of CO is 3.86 cm^{-1} . Calculate the bond length of C–O bond.
- (iv) Show that the rotational spectrum of a rigid diatomic molecule consists of a series of equally spaced lines. 2+2+2+4
- (d) (i) The molar extinction coefficient of phenanthroline complex of iron (II) is $1200 \text{ dm}^3 \text{ mol}^{-1} \text{ cm}^{-1}$ and the minimum detectable absorbance is 0.01. Calculate the minimum concentration of the solution of the given complex that can be detected when taken in a sample cell (cuvette) of path length = 1.00 cm.
- (ii) A 0.003(M) solution of a coloured substance transmits 75% of the incident radiation of 500 nm, when placed in a sample cell of length 1 cm. Calculate the values of (a) molar extinction coefficient (ϵ) and (b) the optical density (O.D.) of a 0.001(M) solution of the same substance taken in the same cell using the same radiation.
- (iii) State and explain Frank-Condon principle.
- (iv) What do you mean by photostationary state? Give an example of quenching of fluorescence. Explain the term 'chemical sensitizer'. 2+(1.5+1.5)+2+(1+1+1)
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